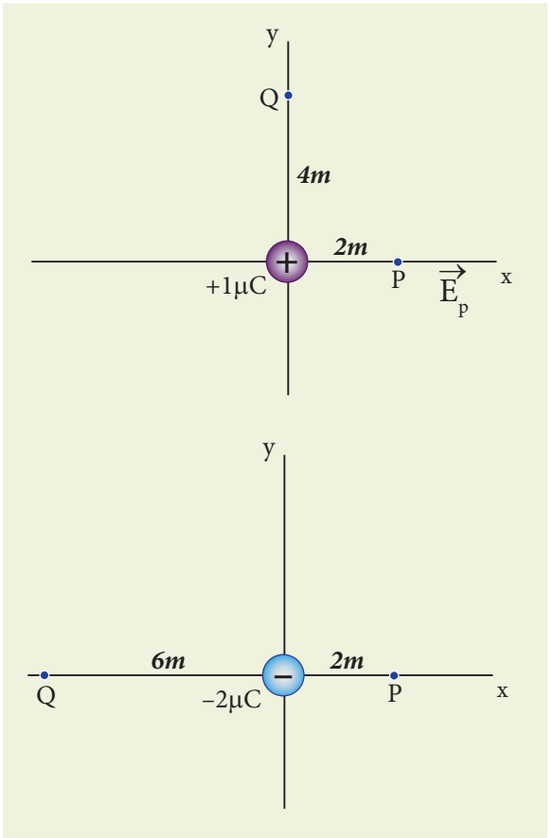
(a) A positive point charge +1 μC is placed at the origin

(b) A negative point charge -2 μC is placed at the origin



Solution

Case (a)

The magnitude of the electric field at point P is

$$E_P = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = \frac{9\times10^9\times1\times10^{-6}}{4}$$

$$= 2.25\times10^3\, NC^{-1}$$

Since the source charge is positive, the electric field points away from the charge. So the electric field at the point P is given by

$$\vec{E}_P = 2.25\times10^3 N\,C^{-1}\,\hat{i}$$

For the point Q

$$\left|\vec{E}_Q\right| = \frac{9\times10^9\times1\times10^{-6}}{16} = 0.56\times10^3\, NC^{-1}$$

Hence $\vec{E}_Q = 0.56\times10^3\,\hat{j}$

Case (b)

The magnitude of the electric field at point P

$$\left|\vec{E}_P\right| = \frac{kq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = \frac{9\times10^9\times2\times10^{-6}}{4}$$

$= 4.5 \times 10^3$ N C^{-1}

Since the source charge is negative, the electric field points towards the charge. So the electric field at the point P is given by

$$\vec{E}_P = -4.5\times10^3\,\hat{i}\; NC^{-1}$$

For the point Q, $\left|\vec{E}_Q\right| = \frac{9\times10^9\times2\times10^{-6}}{36}$

$= 0.5 \times 10^3$ N C^{-1}

$$\vec{E}_R = 0.56\times10^3\hat{i}\; NC^{-1}$$

At the point Q the electric field is directed along the positive x-axis.

